Recommendation and score

Major revision before submission. 72/100 on the rubric used in the previous review, up from 67/100 for v0.10. The new experiments earn credit for testing plausible alternatives. The main unresolved question is whether the edited images support the clinical answers used to score the models.

Dimension	Weight	v0.10	v0.16	Points
Novelty and significance	15	3/5	3/5	9
Methodological soundness	**15**	**3/5**	**4/5**	**12**
Validity of evidence	15	4/5	4/5	12
Statistical analysis	10	4/5	4/5	8
Baselines and fairness	**10**	**3/5**	**4/5**	**8**
Clinical relevance and safety	10	2/5	2/5	4
Reproducibility and transparency	10	4/5	4/5	8
Honesty and limitations	5	4/5	4/5	4
Reporting quality and clarity	5	4/5	4/5	4
References, ethics and declarations	5	3/5	3/5	3

Why the recommendation holds

On 507 main-test cases the selected 4B model claimed an erased lesion in 9.6% of cases, against 41–62% for nine frontier models. On unedited fresh-external pneumothorax positives it detected 0/48. Both results belong in the headline. A claim about reporting behavior is defensible. A claim that an erased finding was clinically absent still depends on expert review.

Scope of verification

I compared the v0.16 manuscript with the supplied Results Summary, dated Results Log and the v0.10 audit (d029d26). That audit independently rescored 134,309 rows across 186 earlier runs. The new comparisons below are author-reported and cross-checked across documents. The current packet contained no new per-view outputs for an independent rerun of C1–C4. Scores are editorial judgments.

What the new experiments establish

C1 • Does the control edit need to match?

The single-seed matched and unmatched arms both claimed an erased lesion in 15.1% of main-test cases. On fresh-external cases the rates diverged (8.3% vs 27.4%), alongside a change in covered-and-correct rate (0.488 vs 0.670). The registered decision was inconclusive. Tool and mask shape changed together, so the comparison cannot isolate either ingredient. Credit the experiment and report the trade-off.

C2 • Does the recipe outperform removal-based preference training?

Across three seeds, fresh-external evidence sensitivity gap (ESG) was 0.381 ± 0.018 for the recipe and 0.355 ± 0.015 for adapted DPO. Seed-specific differences were +0.012, +0.006 and +0.060, with all three intervals spanning zero. On the main test, the recipe claimed fewer erased lesions (13.2% vs 24.2%) but had lower covered-and-correct rate (0.871 vs 0.903). Replace “DPO ties on ESG” with “the study did not establish a fresh-external ESG advantage for the recipe.” Report compute for each adaptation.

C3 and C4 • Do extra training objectives help?

Adding DPO after the recipe raised held-out-tool erased claims from 16.3% to 24.4% in C3. In C4, paired-view consistency training yielded 12.6% against 11.5% with compute-matched continuation. The difference of +1.1 percentage points (95% CI −0.9 to +3.0) missed the primary endpoint. Main-test healthy-image claims also increased from 7.9% to 13.1%. Lower control-edit denial is a secondary benefit. The paired objective did not earn promotion.

Fairness and confirmation

The focused 150-case flagship comparisons add high image-processing settings and free-form replies. These checks strengthen the baseline score under the tested input cap. Describe the actual pixel inputs and settings for each provider. The experiment does not establish equivalence across resolutions.

The original fresh cohorts evaluated the locked v4 model. Later hypotheses used results already observed on these cohorts. The log places the successful C4 pilot before v7 registration and records a healthy-image failure that informed replay. Label C4 as a registered comparison after a pilot on previously examined cases. Reserve independent confirmation for a new patient-disjoint cohort.

Where to check Manuscript §§3.6–3.9, 4.1–4.2, 5.3 and 5.5–5.6. Results Log entries for C1–C4, especially the pilot on 7 November and the three-seed result on 8 November.

What would earn 5/5?

Novelty and significance · 3/5 today · 6 weighted points available

The strongest candidate is a measurement contribution. Removal-based evaluation and preference training already have a close precedent in He et al. (NAACL 2025). MedScope asks a more specific question: does the model withdraw a finding after erasure while retaining it after an unrelated edit? C1–C4 give little reason to claim that a particular recipe uniquely produces this behavior.

1. Demonstrate a decision that changes. Pre-specify a comparison between erasure-only ranking and ranking with the paired control. Name the competing model policies and the smallest difference that would change selection. An always-absent policy, for example, can look perfect on erased-lesion claims while denying every intact control.

2. Test which decision is better. Freeze the selection rule, then compare the selected policies on an untouched external cohort. Use an independent, blinded endpoint such as correct retention of reader-confirmed intact findings and withdrawal of genuinely erased findings. Include unedited sensitivity and healthy-image claims as guardrails. Report the paired difference and interval. A ranking reversal alone earns little credit unless the new ranking makes a better prediction.

3. Show transfer. Repeat the decisive contrast with an unseen editing method or a second clinical source. A material, replicated gain over erasure-only assessment would support 5/5 for measurement novelty. A training-novelty claim instead needs a distinct, reproducible advantage over adapted DPO and an unmatched control at comparable cost.

Methodological soundness · 4/5 today · 3 weighted points available

1. Validate the reference labels. Report blinded specialist judgments for all four image types, including residual or secondary signs, preserved targets, uncertainty and reader agreement. Predefine adjudication. The current 32 erased cases give a two-sided exact 95% upper bound of 10.9% even with zero residual findings. At least 36 independent erasures with zero residuals are needed to clear 10%. Size an extension for a plausible nonzero rate.

2. Rescore with those judgments. The current pack uses different cases for each view type. For a patient-paired ESG analysis, obtain adjudication of both target and control views from the same cases with split-reader blinding. Publish all-case results, confirmed-pair results, exclusions by modality and sensitivity bounds for indeterminate labels.

3. Confirm under a locked protocol. Freeze cohort, endpoints, worthwhile effects, safety margins and model settings. Resample patient groups jointly across edits and tools. Show seed-paired effects separately. Document duplicate exclusions and reproduce the decisive figures from released per-view records. Full methodological credit is compatible with a well-measured negative result.

Priorities before submission

First, establish which cases the benchmark can score

Start with two independent readers qualified for each modality and a third adjudicator for disagreements. Cover thoracic imaging, breast ultrasound, neuroradiology and GI endoscopy as appropriate. This is a proposed design. Reader count cannot compensate for a small case sample. Mask arm identity, record experience, and report agreement on clinical presence as well as edit detectability.

Start validation with localized polyps, nodules and masses. Ask readers whether the finding remains inferable outside the mask. Keep diffuse opacity and pneumothorax as named stress tests until readers establish validity for those erasures. Preserve all unedited positives in the detection analysis. Clinicians should be able to see the 0/48 alongside any improvement on edited images.

Then correct the comparisons a reviewer could misread

Data overlap. One exact image links a validation negative to a fresh-external SIIM positive. Show the fresh-external estimates and intervals after excluding it, including SIIM-specific denominators. Explain why the crosswalk covers 491 main-test images against 507 cases. Pixel similarity cannot certify patient independence across visits, and the 1,834 loose matches remain ambiguous.

Abstract and summary. Primary ESG counts an absent answer or referral. Strict ESG counts explicit absence alone. State which definition each result uses. Correct the summary's six-open-model claim (Qwen2.5-VL-7B had 5.5% erased claims vs 9.6%). Limit p-value language to named endpoints. All 27 grouped primary tests survived Bonferroni, one at $p = 0.001$. Restore the fresh-external utility disadvantage vs Gemini (difference −0.363, 95% CI −0.496 to −0.233).

Publication framing and follow-up work

Venue and utility. Keep one ML-evaluation paper with the training experiments as demonstrations. TMLR is a plausible fit for the scoped contribution. Alongside the assumed utility, give false claims, missed positives and referrals by finding. Where an operating threshold exists, show error versus coverage and compare at fixed coverage. Supply referral burden and sensitivity on all positives so deferral cannot hide misses. TMLR criteria · Geifman & El-Yaniv

Research order. After reader validation, test whether an erase/control pair helps predict or correct errors on new cases. For longer training, first pair pre/post checkpoints by seed and identical view IDs. The cited 16.3% baseline is seed 0 and 11.5% is pooled. A locked training-duration comparison is worthwhile if the paired gain survives without sacrificing clean-image sensitivity. Defer new loss variants until that check is complete.

Submission decision. A reader-validated effect that survives overlap exclusion and an independent comparison would change my recommendation. Complete ethics and consent reporting, the AI-use declaration, figure provenance, model revision and missing table cells. With other rows fixed, full novelty and method credit would bring the score to **81/100**.